

[v2.3] Real analysis

1. Basics

Definition 1.1 (Injective)

A function f is **injective** or **one-to-one** iff:

- $f(a) = f(b) \Rightarrow a = b$, or equivalently
- $a \neq b \Rightarrow f(a) \neq f(b)$

Definition 1.2 (Surjective)

A function $f : A \rightarrow B$ is **surjective** or **onto** iff:

- $f(A) = B$ (its image is its codomain), or equivalently
- for every $b \in B$ there exists $a \in A$ s.t. $f(a) = b$

Theorem 1.3

Composition of relations preserves the following properties:

- function
- injective
- surjective
- continuous

2. Fields and order

2.1. Fields

Definition 2.1.1 (Field)

A **field** $\mathbb{F}$ is any set with relations $+$, $*$, and $=$ which satisfies

- If x and y are in $\mathbb{F}$, so are both $x + y$ and xy .
- $x + y = y + x$ and $xy = yx$.
- $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$.
- $x(y + z) = xy + xz$.
- $w = x$ and $y + z \Rightarrow w + y = x + z$ and $wy = xz$.
- Given $x, y \in \mathbb{F}$, there exists $z \in \mathbb{F}$ s.t. $x + z = y$ (i.e. all addition problems have solutions).

- There exists some number in $\mathbb{F}$ which is not equal to 0. If $x, y \in \mathbb{F}$ s.t. $x \neq 0$, then there exists a real number z s.t. $xz = y$. This z is denoted (y/x) .
- $=$ is an equivalence relation.

2.2. Order

Definition 2.2.1 (Order)

A relation $<$ that satisfies the [total] order axioms:

- **Trichotomy:** $x < y$, $x = y$, xor $x > y$.
- **Transitivity:** If $x < y$ and $y < z$, then $x < z$.

Definition 2.2.2 (Dedekind completeness)

An ordered set is **complete**, **Dedekind-complete**, or has the **least upper bound property** iff:

- every subset of the set that has an upper bound has a supremum, or equivalently
- every subset of the set that has a lower bound has an infimum.

2.3. Ordered fields

Definition 2.3.1 (Ordered field)

A field with an order satisfying

- **Transl.:** If $x < y$, then $x + z < y + z$ for all z .
- **Pos. closure:** If $x > 0$ and $y > 0$, then $xy > 0$.

2.4. Real numbers

Theorem 2.4.1

There is a unique complete ordered field, called $\mathbb{R}$, the **real numbers**.

Theorem 2.4.2 (Approximation Property)

Let S be a nonempty set of reals with a supremum. Then for every real number $a < \sup S$ there exists $x \in S$ with $a < x \leq \sup S$.

Theorem 2.4.3 (Additive Property)

Let A and B be sets of reals with suprema. Consider

$$C := \{a + b : a \in A, b \in B\}.$$

Then,

$$\sup C = \sup A + \sup B.$$

Theorem 2.4.4 (Comparison Property)

Let S and T be two nonempty sets of reals where $s \leq t$ for all $s \in S, t \in T$. Then, if T has a supremum, S also has a supremum and $\sup S \leq \sup T$.

Theorem 2.4.5 (Archimedean property)

The reals are comensural: for any $x, y \in \mathbb{R}$ with $x > 0$, then there is a positive integer n s.t. $nx > y$.

2.5. Topology of the reals

Definition 2.5.1 (Open set)

The following are **open**:

- The empty set
- Open intervals
- Unions of open sets
- Finite intersections of open sets
- The whole space

A set is **closed** if it is the complement of an open set.

The collection of closed sets is closed under finite union and arbitrary intersection.

Definition 2.5.2 (Continuity)

Let $f : X \rightarrow Y$ be a function from topological space X to topological space Y . Then, f is continuous iff $f^{-1}(A) \subset X \Leftrightarrow A$ is open in Y .

2.6. Absolute values

Theorem 2.6.1 (Triangle Inequality)

For any $x, y \in \mathbb{R}$ or $\{x_n\} \subset \mathbb{R}$, or integrable f :

$$|x + y| \leq |x| + |y|$$

$$|x - y| \leq |x| + |y|$$

$$||x| - |y|| \leq |x - y|$$

$$|x_1 + x_2 + \cdots + x_n| \leq |x_1| + |x_2| + \cdots + |x_n|$$

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

3. Sequences

Definition 3.1 (Limit of a sequence)

Let $\{x_n\}$ be a sequence. L is the **limit**, denoted $\lim_{n \rightarrow \infty} x_n$, of the sequence iff for every $\varepsilon > 0$ there exists N s.t. whenever $n > N$, $|x_n - L| < \varepsilon$.

Note - monotonic does not require strictly increasing/decreasing, it can be stable at some points.

Theorem 3.2

A bounded monotonically increasing sequence converges to its supremum.

A bounded monotonically decreasing sequence converges to its infimum.

Theorem 3.3

Every nonempty bounded subset of the reals has an increasing sequence converging to its supremum and a decreasing sequence converging to its infimum.

3.1. Subsequences

Theorem 3.1.1

The following are equivalent:

1. The sequence $\{x_n\}$ converges.
2. The t -tail $\{x_{n+t}\}$ converges for every t .
3. There is some t for which the t -tail converges.

3.2. Limit properties and theorems

Theorem 3.2.1

Let $\{x_n\} \rightarrow x$ and $\{y_n\} \rightarrow y$, s.t. $x_n \leq y_n$ for all n . Then, $x \leq y$.

Theorem 3.2.2 (Squeeze Theorem)

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences s.t. $a_n \leq b_n \leq c_n$ for all n and $a_n \rightarrow L$ and $c_n \rightarrow L$. Then $b_n \rightarrow L$.

Theorem 3.2.3

Let $\{x_n\} \rightarrow x$ and $\{y_n\} \rightarrow y$. Then

1. $\{x_n + y_n\} \rightarrow x + y$.
2. $\{x_n - y_n\} \rightarrow x - y$.
3. $\{x_n y_n\} \rightarrow xy$.
4. $\{x_n/y_n\} \rightarrow x/y$ if $y_n \neq 0$ and $y \neq 0$ for all n .

Theorem 3.2.4

Let $\{x_n\}$ be a sequence and x a number. Then,

$$\{x_n\} \rightarrow x \iff \{x_n - x\} \rightarrow 0.$$

Theorem 3.2.5 (Convergence of geometric series)

Let $c > 0$. Then,

1. $c < 1 \implies \lim_{n \rightarrow \infty} c^n = 0$.
2. $c > 1 \implies \{c^n\}$ is unbounded.

Theorem 3.2.6 (Ratio Test)

Let $\{x_n\}$ be an always-nonzero sequence s.t.

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|} = L.$$

Then,

- $L < 1 \implies \{x_n\} \rightarrow 0$.
- $L > 1 \implies \{x_n\}$ is unbounded.

Theorem 3.2.7

$$\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$$

3.3. Limsup and liminf

Definition 3.3.1 (Limsup and liminf)

Let $\{x_n\}$ be a bounded sequence. For each n , define

$$a_n := \sup\{x_k : k \geq n\}$$

$$b_n := \inf\{x_k : k \geq n\}$$

Thus, a_n is above the n -tail of the sequence, while b_n is below it.

Define

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} a_n$$

$$\liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} b_n.$$

We know that $\limsup x_n \geq \liminf x_n$.

Theorem 3.3.2

If $\{x_n\}$ is bounded, then:

- there is a subsequence $\{x_{n_k}\} \rightarrow \limsup x_n$.
- there is a subsequence $\{x_{m_k}\} \rightarrow \liminf x_n$.

Theorem 3.3.3

Let $\{x_n\}$ be a bounded sequence with $\limsup x_n = \liminf x_n = L$. Then, $\{x_n\} \rightarrow L$.

Theorem 3.3.4

$\{x_n\} \rightarrow L$ iff every subsequence converges to L .

3.4. Bolzano-Weierstrass

Theorem 3.4.1 (Bolzano-Weierstrass for Sequences)

Every bounded sequence of real numbers (or in $\mathbb{R}^n$) has a convergent subsequence.

In topology terms, a subset of $\mathbb{R}^n$ is sequentially compact (every sequence has a convergent subsequence) and compact (every open cover has a finite subcover) iff it is closed and bounded.

Definition 3.4.2 (Open ball)

Let $x \in \mathbb{R}^n$. The ball $B(x, \varepsilon)$ of radius ε about x is defined as

$$B(x, \varepsilon) := \{y \in \mathbb{R}^n : |x - y| < \varepsilon\}.$$

Theorem 3.4.3 (Bolzano-Weierstrass theorem)

Every bounded infinite subset of $\mathbb{R}^n$ has an accumulation point.

Theorem 3.4.4 (Cantor Intersection Theorem)

Let $S_1, S_2, \dots$ be a sequence of bounded closed subsets of $\mathbb{R}^n$ nested as follows: $S_1 \supset S_2 \supset \dots$. Then, the intersection $S_1 \cap S_2 \cap \dots$ is closed and nonempty.

3.5. Cauchy sequences

Definition 3.5.1 (Cauchy sequence)

A sequence $\{x_n\}$ is called **Cauchy sequence** iff for every $\varepsilon > 0$, there exists N s.t. for all $m, n > N$, $|x_n - x_m| < \varepsilon$.

In short, beyond a certain term, all the following terms are arbitrarily close.

Every Cauchy sequence is bounded. A sequence of reals converges iff it is a Cauchy sequence.

4. Series

4.1. Basic convergence tests

Theorem 4.1.1 (n^{th} term test for divergence)

If $\sum x_n$ converges, then $x_n \rightarrow 0$.

Equivalently, if $x_n \not\rightarrow 0$, $\sum x_n$ diverges.

Theorem 4.1.2 (Linearity of series)

Let $\sum x_n = x$ and $\sum y_n = y$. Then, $\sum (x_n + y_n) = x + y$.

For any $c \in \mathbb{R}$, $\sum (cx_n) = cx$.

Theorem 4.1.3 (p-series)

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges iff $p > 1$.

4.2. Absolute convergence

Lemma 4.2.1

If $x_n \geq 0$ for all n , then $\sum x_n$ converges iff the sequence of partial sums is bounded above. Furthermore, each partial sum is less than or equal to the sum.

Definition 4.2.2 (Absolute convergence)

Iff $\sum |x_n|$ converges, then $\sum x_n$ **absolutely converges**. If $\sum |x_n|$ diverges, but $\sum x_n$ converges, then $\sum x_n$ **conditionally converges**.

Theorem 4.2.3 (Comparison test for positive series)

Let $\sum x_n$ and $\sum y_n$ be series with $0 \leq x_n \leq y_n$ for all n . Then:

1. If $\sum y_n$ converges, then $\sum x_n \leq \sum y_n$ ($\sum x_n$ converges).
2. If $\sum x_n$ diverges, so does $\sum y_n$.

4.3. Ratio test and root test

Theorem 4.3.1 (Ratio test)

Let $\sum x_n$ be a series of nonzero terms. Suppose that

$$L := \lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|}$$

exists. Then,

- If $L < 1$, $\sum x_n$ converges absolutely.
- If $L > 1$, $\sum x_n$ diverges.
- If $L = 1$, the test gives no information.

Theorem 4.3.2 (Root test)

Let $\sum x_n$ be a series and $L := \limsup \sqrt[n]{|x_n|}$ (which may be infinite). Then,

- If $L < 1$, $\sum x_n$ converges absolutely.
- If $L > 1$, $\sum x_n$ diverges.
- If $L = 1$, the test gives no information.

4.4. Alternating series

Theorem 4.4.1 (Alternating series test)

Let $x_n \rightarrow 0$ be monotone decreasing. Then, $\sum (-1)^n x_n$ converges.

Theorem 4.4.2

Let $\sum x_n$ be an absolutely convergent series converging to L . Let $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ be a bijection. Then, $\sum x_{\sigma n}$ is also absolutely convergent and converges to L .

In short, if the series is absolutely convergent, we can rearrange the terms and the sum is the same.

Theorem 4.4.3 (Alternating Series Rearrangement Theorem)

Let $\sum x_n$ be a series that satisfies the alternating series test but which is not absolutely convergent.

Then, for any $L \in \mathbb{R}$ there is a rearrangement of the series that sums to L .

4.5. Multiplying series

Definition 4.5.1 (Cauchy product)

The **Cauchy product** of $\sum a_n$ and $\sum b_n$ is defined as

$$\sum_{k=1}^n a_n b_{n+1-k}.$$

Essentially, we collect the terms of the product of $\sum a_n$ with $\sum b_n$ in diagonal strips when the terms are arranged in a grid.

Theorem 4.5.2 (Mertens' theorem)

Let $\sum a_n \rightarrow A$ and $\sum b_n \rightarrow B$. Then if at least one of these is absolutely convergent, then $\sum c_n \rightarrow AB$, where $\sum c_n$ is the Cauchy product of $\sum a_n$ and $\sum b_n$.

Additionally, if both converge absolutely, so does the sum of c_n s.

4.6. Power series

Definition 4.6.1 (Power series)

Given $x_0 \in \mathbb{R}$ and a sequence a_n , we can create the **power series**, a function of x :

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

A power series **converges** iff there exists $x \neq x_0$ that makes the series converge. If $x = x_0$ makes the series converge, then the series converges for any x .

Theorem 4.6.2

A power series either diverges for all $x \neq x_0$, converges absolutely for all x , or there exists a real

number ρ , called the **radius of convergence**, s.t. the series converges absolutely on $(x_0 - \rho, x_0 + \rho)$ and diverges if $|x - x_0| > \rho$.

When $x = x_0 + \rho$ or $x = x_0 - \rho$, this theorem is inconclusive.

5. Limits and continuity

5.1. Limits

Definition 5.1.1 (Accumulation point)

x is a **accumulation point** of S iff

- every ball $B(x, \varepsilon)$ contains a point of S other than x , or, equivalently,
- there is a sequence of points in $S - \{x\}$ converging to x .

Definition 5.1.2 (Limit)

Let S be a subset of $\mathbb{R}$ and c be an accumulation point of S . Let $f : S \rightarrow \mathbb{R}$ be a function.

The **limit** of $f(x)$ as $x \rightarrow c$ is L iff for each $\varepsilon > 0$, there exists a $\delta > 0$ s.t. whenever $x \in S - \{c\}$ and $|x - c| < \delta$, then $|f(x) - L| < \varepsilon$.

Theorem 5.1.3

Let S be a subset of $\mathbb{R}$ and c be an accumulation point of S . Let $f : S \rightarrow \mathbb{R}$ be a function.

Then $f(x) \rightarrow L$ as $x \rightarrow c$ iff for every sequence $\{x_n\}$ in $S - \{c\}$ whose limit is c , the sequence $\{f(x_n)\} \rightarrow L$.

5.2. Continuity

Definition 5.2.1 (Continuity)

Let $S \subset \mathbb{R}$, $c \in S$, and $f : S \rightarrow \mathbb{R}$ be a function.

Then, f is **continuous** at c iff for every $\varepsilon > 0$, there is a $\delta > 0$ s.t. whenever $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.

Theorem 5.2.2

Let $S \in \mathbb{R}$ and $f : S \rightarrow \mathbb{R}$ be a function. Let c be a point in S . Then:

1. If c is not an accumulation point of S , then f is continuous at c .
2. If c is an accumulation point of S , then f is continuous at c iff $\lim_{x \rightarrow c} f(x) = f(c)$.
3. f is continuous at c iff for every sequence $\{x_n\} \in S$ with limit c , $f(x_n) \rightarrow f(c)$.

5.3. Big Continuity Theorems

Definition 5.3.1 (Bounded function)

A function $f : S \rightarrow \mathbb{R}$ is bounded iff its image is bounded.

Lemma 5.3.2

Let $[a, b]$ be a closed finite interval, and f continuous on $[a, b]$. Then f is bounded.

Theorem 5.3.3 (Extreme Value Theorem)

Let $f(x)$ be a continuous function defined on a closed interval $[a, b]$. Then, f achieves an absolute maximum and minimum on $[a, b]$.

Lemma 5.3.4

Let $f(x)$ be continuous on the closed bounded interval $[a, b]$. If $f(a) < 0$ and $f(b) > 0$, then there exists $c \in [a, b]$ s.t. $f(c) = 0$.

Theorem 5.3.5 (Intermediate Value Theorem)

Let f be continuous on closed, bounded $[a, b]$. Let $y \in \mathbb{R}$ s.t. $f(a) < y < f(b)$ or $f(a) > y > f(b)$. Then, there exists $c \in (a, b)$ with $f(c) = y$.

5.4. Uniform continuity

Definition 5.4.1 (Uniform continuity)

Let $S \subset \mathbb{R}$ and $f : S \rightarrow \mathbb{R}$ be a function. For any $\varepsilon > 0$, f is **uniformly continuous** on S iff we can find δ s.t. for all $x, c \in S$ with $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.

This differs from continuity iff the same δ works for every point in S .

Uniform continuity obviously implies continuity.

Theorem 5.4.2

If $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function defined on a closed bounded interval, it is uniformly continuous.

Lemma 5.4.3

If $f : S \rightarrow \mathbb{R}$ is uniformly continuous and $\{x_n\}$ is a Cauchy sequence in S , then $\{f(x_n)\}$ is a Cauchy sequence.

Theorem 5.4.4

Let $f : (a, b) \rightarrow \mathbb{R}$. Then f is uniformly continuous iff it has a right-handed limit at a , a left-handed limit at b , and the extension of f to the closed interval $[a, b]$ is continuous.

5.5. Lipschitz continuity

Definition 5.5.1 (Lipschitz continuity)

Let $f : S \rightarrow \mathbb{R}$ be a function. If there exists $K \in \mathbb{R}$ s.t. for every $x, y \in S$, $|f(x) - f(y)| < K|x - y|$ then f is **Lipschitz continuous**.

Theorem 5.5.2

Lipschitz continuity implies uniform continuity.

6. Derivatives

Definition 6.1 (Derivative)

Let $f(x)$ be a real-valued function defined on an open interval (a, b) , and let $x_0 \in (a, b)$. The **derivative** of f at x_0 is

$$\frac{df}{dx}(x_0) := f'(x_0) := \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

Iff this limit is defined, the function is **differentiable** at x_0 .

Theorem 6.2 (Continuity theorem)

Let f be defined on (a, b) with $x_0 \in (a, b)$. Then, f is differentiable at x_0 iff there is a function f^* (which depends on f and on x_0) s.t.

$$f(x) = f(x_0) + f^*(x)(x - x_0)$$

with f^* defined on (a, b) and continuous at x_0 , with $f^*(x_0) = f'(x_0)$.

If f is continuous, f^* can be defined as

$$f^*(x) := \begin{cases} \frac{f(x) - f(x_0)}{x - x_0} & \text{if } x \neq x_0 \\ f'(x_0) & \text{if } x = x_0. \end{cases}$$

Theorem 6.3 (Sum, product, and quotient rules)

Let f and g be functions defined on an interval (a, b) , both differentiable at the point $x_0 \in (a, b)$. (In particular, the derivatives there cannot be infinite.) Then, the following derivatives exist and take on the following values:

- $(f \pm g)'(x_0) = f'(x_0) \pm g'(x_0)$.
- $(f * g)'(x_0) = f(x_0)g'(x_0) + f'(x_0)g(x_0)$.
- $(f/g)'(x_0) = \frac{g(x_0)f'(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}$.

Theorem 6.4 (Chain rule)

Let $f : (a, b) \rightarrow \mathbb{R}$ and $g : f(a, b) \rightarrow \mathbb{R}$. Let $x_0 \in (a, b)$ with f differentiable at x_0 and $f(x_0)$ contained in the interior of $f(a, b)$, and let g be differentiable at $f(x_0)$. Then, $(g \circ f)(x)$ is differentiable at x_0 , and

$$(g \circ f)'(x_0) = g'(f(x_0))f'(x_0),$$

or equivalently,

$$\left. \frac{dg}{dx} \right|_{x_0} = \left. \frac{dg}{df} \right|_{f(x_0)} * \left. \frac{df}{dx} \right|_{x_0}.$$

6.1. Mean value theorem

Theorem 6.1.1

Let $f'(c) > 0$ and f continuous in some interval about c . Then, there is a $\delta > 0$ s.t.:

- $c < x < c + \delta \implies f(c) < f(x)$
- $c - \delta < x - c \implies f(x) < f(c)$

Equivalently, if the derivative is positive, then the function is increasing at that point.

Similarly, if the derivative is negative, then the function is decreasing at that point.

Corollary 6.1.1.1

If $f(c)$ is a local min or local max, then $f'(c) = 0$.

Definition 6.1.2 (Critical point)

A point where the derivative of the function is zero.

Theorem 6.1.3 (Rolle's theorem)

Let $f(x)$ be continuous on $[a, b]$ and differentiable on (a, b) , s.t. $f(a) = f(b)$. Then, there exists $c \in (a, b)$ for which $f'(c) = 0$.

Theorem 6.1.4 (Mean value theorem)

Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then, there exists some point $c \in (a, b)$ for which

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem 6.1.5 (Cauchy's mean value theorem)

Let $f(t)$ and $g(t)$ be two functions continuous on $[a, b]$ and differentiable on (a, b) . Then, there exists some point $c \in (a, b)$ for which

$$f'(c)(g(b) - g(a)) = g'(c)(f(b) - f(a)).$$

Theorem 6.1.6

Let $f(x)$ be continuous on $[a, b]$ and differentiable on (a, b) . Then:

- $f'(x) > 0$ for all $x \in (a, b) \iff f$ is strictly increasing
- $f'(x) < 0$ for all $x \in (a, b) \iff f$ is strictly decreasing
- $f'(x) = 0$ for all $x \in (a, b) \iff f$ is constant

Let $g(x)$ also be continuous on $[a, b]$ and differentiable on (a, b) . Then:

- $g'(x) = f'(x)$ for all $x \in (a, b) \iff f(x) - g(x)$ is constant

Theorem 6.1.7

Let $f(x)$ be continuous on $(a, b]$ and differentiable on (a, b) , with $\lim_{x \rightarrow b^-} f'(x) = L$. Then, f is left-hand differentiable at b , and $f'_-(b) = L$.

A similar theorem exists for right-hand derivatives.

Theorem 6.1.8 (Darboux theorem)

Let $f(x)$ be a function that is differentiable on $[a, b]$. Let y be a value s.t. $f'(a) < y < f'(b)$ or $f'(a) >$

$y > f'(b)$. Then there is some $c \in (a, b)$ s.t. $f'(c) = y$.

Corollary 6.1.8.1

- f' cannot change signs without going through zero.
- If $f'(x) \neq 0$ for any x , then f is monotonic.
- If f' itself is monotonic then it must be continuous:
 - the only kinds of discontinuities of monotonic functions are jumps
 - derivatives can't make jumps

6.2. Taylor series

Theorem 6.2.1 (Taylor's formula with the Lagrange form of the remainder)

Let $f : (a, b) \rightarrow \mathbb{R}$ have $n + 1$ derivatives on the interval (a, b) , and let x_0 be a point in (a, b) . Then, for any point y in (a, b) **Taylor's formula** states that

$$f(y) = p_n(y) + R_n(y; x_0).$$

We define the **n th Taylor polynomial centered at x_0** as:

$$p_n(x) := \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

The **Taylor series** is the limit of the Taylor polynomials.

We define the **Lagrange form of the remainder** as:

$$R_{n(x; x_0)} := \frac{f^{(n+1)}(c)(y - x_0)^{n+1}}{(n + 1)!}$$

for some c between y and x_0 [this theorem does not specify which c this is].

7. Integrals

7.1. Partitions

Definition 7.1.1 (Partition)

A **partition** P of $[a, b]$ is a finite set of points, $\{x_0, x_1, \dots, x_n\}$ with $a = x_0 < x_1 < \dots < x_n = b$. The k th interval of the partition is $[x_{k-1}, x_k]$, with width $\Delta x_k := x_k - x_{k-1}$.

Definition 7.1.2 (Norm/mesh)

The **norm** or **mesh** of a partition is the width of its largest interval, denoted

$$\|P\| := \max\{\Delta x_k : k \leq n\}.$$

Definition 7.1.3 (Refinement)

$\tilde{P}$ is a **refinement** of P iff $P \subseteq \tilde{P}$.

7.2. Darboux integral

Definition 7.2.1 (Darboux sum)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and P a partition of $[a, b]$. Then, define

$$m_k := \inf\{f(x) : x_{k-1} \leq x \leq x_k\}$$

$$M_k := \sup\{f(x) : x_{k-1} \leq x \leq x_k\}.$$

Define the **upper** and **lower Darboux sums**, respectively, as

$$L(P, f) := \sum_{k=1}^n m_k \Delta x_k$$

$$U(P, f) := \sum_{k=1}^n M_k \Delta x_k.$$

Definition 7.2.2 (Upper and lower Darboux integrals)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Then, define the **upper** and **lower Darboux integrals** as follows:

$$\int_a^b f(x)dx := \inf\{U(P, f) : P \text{ is a partition of } [a, b]\}$$

$$\int_a^b f(x)dx := \sup\{L(P, f) : P \text{ is a partition of } [a, b]\}.$$

Definition 7.2.3 (Darboux integrability)

If $f : [a, b] \rightarrow \mathbb{R}$ is bounded, and $\int_a^b f(x)dx = \overline{\int_a^b f(x)dx}$, then we say f is **Darboux-integrable** ($f \in \mathcal{R}[a, b]$), and we define $\int_a^b f(x)dx$ to be the common value of the lower and upper Darboux integral.

7.3. Riemann integral

Definition 7.3.1 (Riemann sum)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and P a partition of $[a, b]$.

For each interval $[x_{i-1}, x_i]$, select any t_i in that interval. Then, the **Riemann sum** of f under this selection of t_i is

$$\sum_{i=1}^n f(t_i) \Delta x_i.$$

Definition 7.3.2 (Riemann integrability)

f is **Riemann-integrable** ($f \in \mathcal{R}[a, b]$) iff there exists some number I s.t. for every $\varepsilon > 0$, there exists a partition P_0 s.t. for every refinement P of P_0 and every choice of t_i on P , the corresponding Riemann sum is within ε of I .

Theorem 7.3.3

Riemann integrability is equivalent to Darboux integrability.

7.4. Integrability

Theorem 7.4.1

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Then, f is Darboux-integrable iff for every $\varepsilon > 0$, there exists a partition P s.t. $U(P, f) - L(P, f) < \varepsilon$.

Theorem 7.4.2

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and monotonic. Then, $f \in \mathcal{R}[a, b]$.

7.5. Core properties of the integral

Theorem 7.5.1 (Additivity)

Let $a < b < c$ and $f : [a, c] \rightarrow \mathbb{R}$ be a bounded function. Then,

$$f \in \mathcal{R}[a, c] \iff f \in \mathcal{R}[a, b] \text{ and } f \in \mathcal{R}[b, c],$$

and if f is integrable on the whole interval then

$$\int_a^c f = \int_a^b f + \int_b^c f.$$

Theorem 7.5.2 (Linearity)

Let $f, g \in \mathcal{R}[a, b]$, and α any real constant. Then,

$$\int_a^b \alpha f(x) + g(x)dx = \alpha \int_a^b f(x)dx + \int_a^b g(x)dx.$$

Theorem 7.5.3 (Monotonicity)

Let f and g be bounded functions on $[a, b]$. Assume $f(x) \leq g(x)$ for all $x \in [a, b]$. Then, $\int_a^b f < \int_a^b g$ and $\int_a^b f < \int_a^b g$. If both are in $\mathcal{R}[a, b]$ then $\int_a^b f < \int_a^b g$.

7.6. Additional properties of the integral

Theorem 7.6.1

Let $f \in [a, b]$. Let $f(x) \in [m, M]$ for all $x \in [a, b]$. Then,

$$m(b-a) \leq \int_a^b f(x)dx \leq M(b-a).$$

Theorem 7.6.2 (Integral of a constant)

$$\int_a^b cdx = c(b-a).$$

Theorem 7.6.3 (Integral of a step function)

Let

$$f(x) = \begin{cases} r & \text{for } x < c \\ s & \text{for } x = c \\ t & \text{for } x > c \end{cases}$$

with $c \in [a, b]$ and $f : [a, b] \rightarrow \mathbb{R}$.

Then,

$$\int_a^b f(x)dx = r(c-a) + t(b-c).$$

Definition 7.6.4

$$\int_a^b f(x)dx := - \int_b^a f(x)dx.$$

Theorem 7.6.5 (Integration by parts)

If u and v are integrable and have continuous derivatives,

$$\int_a^b u(x)v'(x)dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)dx.$$

7.7. Integrals of continuous functions

Lemma 7.7.1

A bounded function that has only finitely many discontinuities is integrable.

Lemma 7.7.2

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Let $\{a_n\} \rightarrow a$ and $\{b_n\} \rightarrow b$ be sequences in $[a, b]$, with $a_n < b_n$ for all n . Then, if $f \in \mathcal{R}[a_n, b_n]$ for all n , then $f \in \mathcal{R}[a, b]$ and $\int_a^b f = \lim_{n \rightarrow \infty} \int_{a_n}^{b_n} f$.

Essentially, bounded functions need only be integrable inside the interval to be integrable on the entire interval.

Theorem 7.7.3

Continuous functions are integrable.

7.8. Fundamental theorem of calculus

Theorem 7.8.1 (First Fundamental Theorem of Calculus)

Let $F : [a, b] \rightarrow \mathbb{R}$ be continuous and differentiable* on (a, b) , with $F' \in \mathcal{R}[a, b]$. Then,

$$\int_a^b F'(x) dx = F(b) - F(a).$$

* F may have finitely many points where it is not differentiable. It must be continuous over the whole interval.

Theorem 7.8.2 (Second Fundamental Theorem of Calculus)

Let $f \in \mathcal{R}[a, b]$. Define $F(x) := \int_a^x f$. Then, F is continuous on $[a, b]$, and if f is continuous at c , then F is differentiable at c and $F'(c) = f(c)$.

Theorem 7.8.3 (u-substitution)

Let $f : [c, d] \rightarrow \mathbb{R}$ be continuous and $g : [a, b] \subset [c, d]$ continuously differentiable. Then,

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Note that in this case, $u = g(x)$ and $du = g'(x)dx$.

8. Sequences of functions

8.1. Pointwise convergence

Definition 8.1.1 (Pointwise convergence)

Let $\{f_1, f_2, \dots\}$ be a sequence of functions. Let S be the set of all points x for which the sequence $\{f_1(x), f_2(x), \dots\}$ converges. Then, S is the domain of the limit function

$$f(x) := \lim_{n \rightarrow \infty} f_n(x).$$

$\{f_n\}$ *converges pointwise* to f on S .

8.2. Uniform convergence

Definition 8.2.1 (Uniform convergence)

Let $\{f_n\}$ be a sequence of functions. $\{f_n\}$ *converges uniformly* to f on S iff for every $\varepsilon > 0$ there exists N s.t. $n > N$ implies $|f(x) - f_n(x)| < \varepsilon$ for all $x \in S$.

Definition 8.2.2

Let $\{f_n\}$ be a sequence of functions on S . Let $M > 0$ be a constant so that $|f_n(x)| \leq M$ for all x and n . Then, $\{f_n\}$ is *uniformly bounded* by M on S .

Theorem 8.2.3

Let $f_n \rightarrow f$ uniformly on S and let each f_n be bounded. Then, f is bounded and the sequence $\{f_n\}$ is uniformly bounded on S .

Definition 8.2.4 (Uniform norm)

Let f be a bounded function on domain S . Then, define the *uniform norm* or *infinity norm* of f on S is

$$\|f\|_\infty = \|f\|_u := \sup\{|f(x)| : x \in S\}.$$

Theorem 8.2.5

A sequence of bounded functions f_n converges uniformly to f iff

$$\lim_{n \rightarrow \infty} \|f_n - f\|_\infty = 0.$$

Theorem 8.2.6

Let $\{f_n\}$ be a sequence of functions. Then there exists a function f s.t. $f_n \rightarrow f$ uniformly iff the sequence satisfies the **Cauchy condition**: for every $\varepsilon > 0$ there exists N s.t. if $m, n > N$, then $|f_m(x) - f_n(x)| < \varepsilon$ for all x .

Definition 8.2.7

A series converges uniformly iff its sequence of partial sums converges uniformly.

Theorem 8.2.8 (Weierstraß M-test)

Let M_n be a sequence of nonnegative numbers s.t. $|f_n(x)| \leq M_n$ for all x and n . Then, $\sum f_n$ converges uniformly if $\sum M$ converges.

8.3. Continuity, integrability, and differentiability of limit functions

Theorem 8.3.1 (Interchanging limits)

When $f_n \rightarrow f$ uniformly, then

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} \lim_{n \rightarrow \infty} f_n(x) \\ &= \lim_{n \rightarrow \infty} \lim_{x \rightarrow a} f_n(x) = \lim_{n \rightarrow \infty} f_n(a) = f(a). \end{aligned}$$

Theorem 8.3.2

Let $\{f_n\}$ be a sequence of continuous functions $S \rightarrow \mathbb{R}$, where S is some domain. If the sequence converges uniformly to a function f , then f is also continuous.

Theorem 8.3.3

Let $\{f_n\}$ be a sequence of integrable functions on $[a, b]$ that converges uniformly to f . Then f is integrable, and

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

Theorem 8.3.4

Let each term of $\{f_n\}$ be differentiable at each point of (a, b) and assume that these derivatives converge $f'_n \rightarrow g$ uniformly on (a, b) . Further, assume that there is at least one point $x_0 \in (a, b)$ s.t. $\{f_n(x_0)\}$ converges. Then,

- There is a function f s.t. $f_n \rightarrow f$ uniformly on (a, b) , and
- For each point in x in (a, b) , the derivative $f'(x)$ exists and equals $g(x)$.

Theorem 8.3.5

Upon its interval of convergence:

- The power series of a function converges uniformly to a continuous integrable
- The integral of the sum of the series is the sum of the integrals of the terms
- The derivatives of the power sums converge uniformly
- The partial sums of the series add to $f(x_0)$ at x_0
- Can differentiate a power series term-by-term to obtain the derivative of the function to which it converges.

9. Picard's theorem

Definition 9.1 (Lipschitz in the second variable)

$F : \mathbb{R}^2 \rightarrow \mathbb{R}$ is **Lipschitz in the second variable** iff there exists $L \in \mathbb{R}$ s.t.

$$|F(x, y) - F(x, z)| \leq L|y - z|$$

for all $y, z \in J$ and $x \in I$.

Theorem 9.2 (Picard's theorem)

Let $I, J \in \mathbb{R}$ be closed bounded intervals with $(x_0, y_0) \in I^\circ \times J^\circ$.

If $F : I \times J \rightarrow \mathbb{R}$ is continuous and Lipschitz in the second variable, then there exists $h > 0$ s.t. $[x_0 - h, x_0 + h] \subset I$ and a unique differentiable function $f : [x_0 - h, x_0 + h] \rightarrow J$ s.t.

$$f'(x) = F(x, f(x)) \quad f(x_0) = y_0.$$

In short, if F is continuous and Lipschitz in the second variable, the initial value problem has a unique solution on some interval around x_0 .

The unique solution can be constructed via the following procedure:

1. Define $f_0(x) := y_0$.
2. For each $k \in \mathbb{N}^+$, define the ***k*th Picard iterate**:

$$f_k(x) := y_0 + \int_{x_0}^x F(t, f_{k-1}(t)) dt$$

3. The sequence of Picard iterates converges uniformly to the solution f .

10. Lebesgue integrability condition

10.1. Oscillation

Definition 10.1.1 (Oscillation)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $S \subset \mathbb{R}$. The **oscillation** of f on S is defined as

$$\omega_f(S) := \sup\{f(S)\} - \inf\{f(S)\}.$$

If the supremum or infimum does not exist, it is defined as $+\infty$.

The **oscillation** of f at the point $x \in S$ is defined as

$$\lim_{\varepsilon \rightarrow 0} \omega_f(S \cap (x - \varepsilon, x + \varepsilon)).$$

Theorem 10.1.2

f is continuous at x_0 iff the oscillation of f at x_0 is zero.

10.2. Measure

Definition 10.2.1 (Measure)

A **measure** is a function that takes a subset S of the reals, and returns its “size” denoted $|S|$, following the following properties:

- $0 \leq |S| \leq \infty$.
- $|\emptyset| = 0$.
- $S' := \{x + a : x \in S\} \implies |S'| = |S|$.
- $|(0, 1)| = |[0, 1]| = 1$.
- **Countably additive**: If $\{S_i\}$ is a countable collection of disjoint sets, $|\cup S_i| = \sum_i |S_i|$.

10.2.1. Lebesgue outer measure

Definition 10.2.2 (Lebesgue outer measure)

Cover the set with a countable number of disjoint open intervals, and sum their totals. Take the infimum of all such coverings.

$$\lambda^*(S) := \inf\left\{\sum (b_i - a_i) \mid \{(a_i, b_i)\}_{i \in \mathbb{Z}^+} \text{ is such a cover}\right\}.$$

Theorem 10.2.3

- Every countable subset of $\mathbb{R}$ has [Lebesgue outer] measure 0.
- A subset of a measure 0 set also has measure 0.
- Any countable union of measure 0 sets has measure 0.

10.3. Lebesgue integrability condition

Theorem 10.3.1 (Lebesgue Integrability Condition)

$f : [a, b] \rightarrow \mathbb{R}$ is Riemann-integrable iff it is bounded and its set of discontinuities has measure 0.

Corollary 10.3.1.1

Let $f : [a, b] \rightarrow [c, d]$ be integrable and $g : [c, d] \rightarrow \mathbb{R}$ continuous. Then, $g \circ f$ is integrable.

11. Counterexamples & Constructions

11.1. Pathological Functions

P1. Dirichlet function: $D(x) = 1$ if $x \in \mathbb{Q}$, 0 if $x \notin \mathbb{Q}$. *Properties:* bounded; nowhere continuous (every interval contains both rationals and irrationals); NOT integrable ($U(P, D) = 1$, $L(P, D) = 0$ for any partition P); can be written as $D(x) = \lim_{m \rightarrow \infty} (\lim_{n \rightarrow \infty} \cos^{2n}(m! \pi x))$.

P2. Thomae's function (Modified Dirichlet): $f(\frac{p}{q}) = \frac{1}{q}$ (lowest terms), $f(\text{irrational}) = 0$. *Properties:* bounded; continuous at every irrational (hence almost everywhere); discontinuous at every rational; Riemann integrable with $\int_0^1 f = 0$; discontinuities are countable (measure 0). Shows the Lebesgue Integrability Condition in action.

P3. $f(x) = x \sin(\frac{1}{x})$, $f(0) = 0$. *Properties:* continuous everywhere; uniformly continuous on $[0, 1]$; NOT Lipschitz at 0 (oscillates arbitrarily steeply); NOT differentiable at 0.

P4. $f(x) = x^2 \sin(\frac{1}{x})$, $f(0) = 0$. *Properties:* differentiable everywhere; $f'(0) = 0$; $f'(x) = 2x \sin(\frac{1}{x}) - \cos(\frac{1}{x})$ for $x \neq 0$; $f' \nrightarrow 0$ as $x \rightarrow 0$, so f' is NOT continuous at 0 (f is differentiable but $f' \notin C^1$).

P5. $f_{n(x)} = x^n$ on $[0, 1]$. Pointwise limit: $f(x) = 0$ for $x < 1$, $f(1) = 1$ (discontinuous). Convergence is NOT uniform: $\|f_n - f\|_\infty = 1$ for all n . Shows the uniform limit of continuous functions need not hold under pointwise convergence.

P6. Weierstraß function: $W(x) = \sum_{n=0}^{\infty} (\frac{1}{2})^n \cos(3^n \pi x)$. Continuous everywhere, nowhere differentiable ($\|f_n - f\|_\infty \rightarrow 0$ by the Weierstraß M -test, but derivatives blow up). Limit of smooth functions can fail to be smooth.

P7. $f(x) = \sqrt{x}$ on $[0, 1]$. Uniformly continuous (compact domain and continuous). NOT Lipschitz at 0: $\frac{|\sqrt{x}-0|}{|x-0|} = \frac{1}{\sqrt{x}} \rightarrow \infty$. Shows uniform continuity $\nRightarrow$ Lipschitz.

P8. $f(x) = \frac{1}{x}$ on $(0, 1)$. Continuous. NOT uniformly continuous (δ must depend on position; Cauchy sequences map to non-Cauchy sequences). NOT bounded. Shows $f : (a, b) \rightarrow \mathbb{R}$ is uniformly continuous if and only if it extends continuously to $[a, b]$.

P9. $f(x) = |x|$. Uniformly continuous; Lipschitz (constant $K = 1$); NOT differentiable at 0. Shows Lipschitz $\nRightarrow$ differentiable.

P10. If $|f(x) - f(y)| \leq (x - y)^2$ for all x, y , then $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = 0$ everywhere, so f is constant.

P11. The alternating harmonic series $\sum \frac{(-1)^n}{n}$ converges (by the alternating series test) but NOT absolutely (since $\sum \frac{1}{n}$ diverges). Any real number can be achieved by rearrangement (Riemann Rearrangement Theorem).

11.2. Property Constructions (No Proof)

• **Continuous, not uniformly continuous:** $f(x) = x^2$ on $\mathbb{R}$; or $\sin(\frac{1}{x})$ on $(0, 1]$.

• **Uniformly continuous, not Lipschitz:** $f(x) = \sqrt{x}$ on $[0, 1]$.

• **Lipschitz, not differentiable:** $f(x) = |x|$ (constant $K = 1$).

• **Pointwise convergent, not uniformly convergent:** $f_{n(x)} = x^n$ on $[0, 1]$.

• **Uniformly convergent, but derivatives may not converge:** $f_{n(x)} = \frac{\sin(nx)}{n} \rightarrow 0$ uniformly, but $f_{n'}(x) = \cos(nx) \nrightarrow 0$.

• **Bounded, not integrable:** Dirichlet function $D(x)$.

• **Integrable, not continuous (even with dense discontinuities):** Thomae's function.

• **Integrable f , injective g , but $f \circ g$ not integrable:** Take $f = \text{Thomae}$ (discontinuous on $\mathbb{Q}$) and g a bijection mapping irrationals to rationals (e.g. Cantor-based); $f \circ g$ is 0 on irrationals, $\frac{1}{q}$ on images of rationals — can be made non-integrable.

• **Sequence with $\limsup \neq \liminf$:** $\{(-1)^n\}$: $\limsup = 1$, $\liminf = -1$.

• **Limit comparison test failure:** $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$: ratio $\rightarrow \infty$, both diverge/converge differently — shows the limit comparison test requires $0 < L < \infty$.

11.3. Quick Reference: Implications

- Lipschitz $\Rightarrow$ Uniformly continuous $\Rightarrow$ Continuous
- Differentiable $\Rightarrow$ Continuous
- Uniformly convergent $\Rightarrow$ Pointwise convergent
- Absolutely convergent $\Rightarrow$ Convergent
- Closed + bounded $\Leftrightarrow$ Sequentially compact
- Continuous $\Rightarrow$ Integrable

None of these reverse in general.